DSA06 - ACTIVITY 5

NORMALITY DETECTION USING Q-Q PLOTS

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Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). For every chart, decide whether the data are normal and name the shape from the list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

SOLUTION:

Part A :

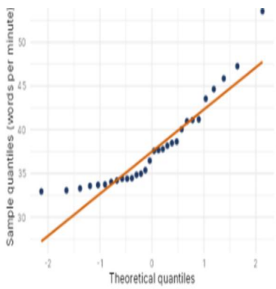
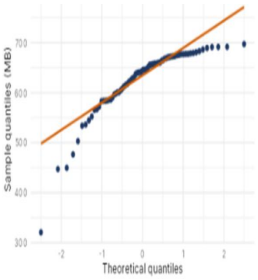
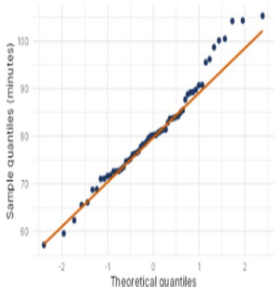
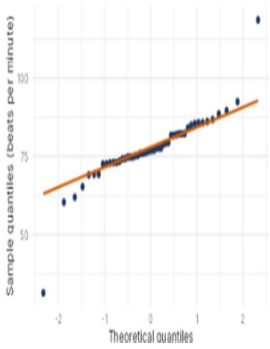
Chart	Chart Image	Chart Type (Data)	Normal?	Shape	Evidence in the plot
Chart 1		Typing speed (words per minute), n = 30	**No**	**Right-skewed**	Dots form a convex (U-shaped) curve. Many values are bunched at about 33 to 36 wpm, below the line. The upper end bends away above the line (about 47 to 53 wpm), showing a long upper tail.
Chart 2		Mobile data used per day (MB), n = 80	**No**	**Left-skewed**	Dots form a concave (arch-shaped) curve. The lowest value (about 320 MB) is far below the line, while the upper end flattens out at about 690 MB below the line, showing a long lower tail.
Chart 3		Lab experiment duration (minutes), n = 60	**Yes**	**Normal**	Dots lie along the red line from end to end with only small random wiggles; no systematic curve, S-shape or isolated point.

Chart	Chart Image	Chart Type (Data)	Normal?	Shape	Evidence in the plot
Chart 4		Resting pulse (beats per minute), n = 50	**No (outliers)**	**Normal with outlier(s)**	Most dots (about 60 to 90 bpm) sit close to the line, but two isolated points are far away: one very low (about 30 bpm) and one very high (about 116 bpm).

Part B

B1. Chart 1 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods. [2]

Ans: Chart 1 is right-skewed: the dots curve upward and the upper tail (about 47 to 53 wpm) stretches far above the line. The remedy is a log transform of the typing speed (a square-root transform is a milder alternative), which compresses the long upper tail and pulls the data towards a normal shape. The Q-Q plot is then redrawn to check that the dots follow the line. If it does not help, use robust methods (median and IQR) instead of mean and SD.

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

Ans: ggplot(df, aes(sample = value)) + stat_qq() + stat_qq_line()

- stat_qq(): draws the dots (sample quantiles against theoretical normal quantiles).
- stat_qq_line(): draws the red reference line showing the perfect normal fit.

Summary

Chart	Normal / Not normal | Shape
Chart 1: Typing speed (words per minute)	Not normal | Right-skewed
Chart 2: Mobile data used per day (MB)	Not normal | Left-skewed
Chart 3: Lab experiment duration (minutes)	Normal | Normal
Chart 4: Resting pulse (beats per minute)	Not normal | Normal with outlier(s)

Conclusion

A Q-Q plot compares the data with a perfect normal distribution. Dots that follow the red line mean the data are normal, a smooth curve means skewness, and isolated far-away points mean outliers. When data are skewed or contain outliers, a transformation (such as log) or robust measures (median and IQR) should be tried before relying on mean-and-SD methods.